

BIOL550 Mouse RNA-seq Pipeline Architecture

Standalone stage modules, explicit checkpoints, and reusable handoffs across collection, cleaning, preparation, and analysis

Pipeline principle: each stage owns its own module logic, evidence checkpoint, and handoff artifact

1. DATA COLLECTION

Study intake + metadata free

- SRA / BioProject / GEO records
- RunInfo snapshot + accession link
- family manifest + design table

2. DATA CLEANING

Trimming toolbench

- FASTX baseline branch
- fastp default cleanup branch
- cutadapt targeted-fallback branch

3. DATA PREPARATION

Reference selection module

- GRCh38 primary assembly
- Ensembl v115 annotation
- reference freeze for reproducibility

4. DATA ANALYSIS & INTERPRETATION

DE standalone module

- DESeq2 interaction design
- ~ side_class * geno_class
- contrast inventory fixed before interpretation

Parallel acquisition module

- prefetch + fasterq-dump wrapper
- worker-based raw-stage execution
- resumable logs + run-level checkpoints

Validation module

- manual-first validation of 1 run
- scripted automation for n-1 remaining
- custom comparison workflow

STAR implementation module

- genomeGenerate / index build
- paired-end alignment
- GeneCounts in same run

Interpretation control module

- PCA-first structure review
- sample-distance checks
- bend-point narrowing of large DE sets

Raw QC module

- FastQC per run
- raw MultiQC aggregation
- raw QC bundle copied locally

Post-trim QC corroborator

- trimmed FastQC per run
- trimmed MultiQC summary
- GC bell-shape + severity-deliverables

Alignment review + count handoff

- Log.final.out parsing
- mapping / noFeature / ambiguous
- reverse-stranded count matrix (count)

GO / follow-up module

- g:Profiler: GO:BP / KEGG / Reactome
- shared-gene chart / tree / network views
- anchor genes + report / paper figures

Handoff: raw FASTQ pairs + run metadata

Handoff: cleaned FASTQs + per-run QC reports

Handoff: BAMs + STAR logs + count matrix

Handoff: DE tables + bend-point follow-up sets + enrichment summaries

Collection + Cleaning preserve both per-run evidence and aggregated MultiQC summaries for raw vs. cleaned states.

Preparation is explicitly split into reference choice, index generation, and alignment/count handoff to keep STAR reproducible.

Analysis stays modular: DE, bend-point narrowing, and GO follow-up are separated so interpretation does not drift.